

Stems and Branches: From One Tree to Twenty-Two Characters

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2026-10-04 · 预印本 (preprint) · 未经同行评议 · v2

DOI: 10.5281/zenodo.22949963

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Abstract

Stems and branches (干支) are usually met as a tool: they name the year in the four-pillar chart, mark the months, and divide the day into double-hours, and yet they are rarely explained in themselves. This note supplies that explanation. It begins with the two characters, which originally meant "trunk" and "branch" — the main stem of a tree and its bough — and with Cai Yong's account, in the *Yueling zhangju*, of how the system was made to name days and months. The ten heavenly stems are then read in order as the life of a single plant, from the cracking of the shell (jia) to the storing of the seed (gui), with the source line for each step, chiefly from the *Shiji*, "Treatise on Pitch Pipes", the *Shuowen jiezi* and the *Shiming*. The twelve earthly branches are read instead as the coming and going of qi, with zi in the eleventh month and wu at the summer solstice as its two pivots, and the *Hanshu*, "Treatise on Pitch Pipes and the Calendar", stringing the twelve together. Two rules follow: yin and yang by odd and even position, the five phases by direction. The note explains why the branches carry two numberings that do not conflict, and why chen, xu, chou and wei are the four points of handover. It closes by dating the pairing of the branches with animals, from the *Lunheng* back to three excavated daybooks. The note is descriptive; it contains no predictive claims.

Keywords: stems and branches; heavenly stems; earthly branches; yin and yang; five phases; zodiac animals; *Lunheng*; excavated daybooks

1. The stem is the trunk, the branch is the bough

In the preceding ten notes the stems and branches have been used as a tool several times. They appear everywhere — in the year pillar, the month establishment, the double-hour — and not once have they been explained for what they are.

This is not a detour. The year is read from lichun, the month from the solar term, the hour from the sun. If the measuring rod has not first been aligned, the symbols will be of no use however well one knows them. Align the rod first; then learn the symbols.

Jia, yi, bing, ding — you can read them, you can write them; you learned them in primary school. You have probably also heard it said that jia, yi, bing, ding stand for "so on", the equivalent of A, B, C, D.

They were not originally meant for ordering anything.

The two characters gan zhi (干支) were originally gan zhi (幹枝): the trunk of a tree and its boughs.

Cai Yong of the Eastern Han, in the *Yueling zhangju*, recorded where the system came from:

命大桡探五行之情，占斗纲所建，于是始作甲乙以名日，谓之干；作子丑以名月，谓之支。干支相配以成六旬。（He charged Da Rao to investigate the disposition of the five phases, and to divine what the handle of the Dipper established; thereupon he first made jia and yi to name the days, calling them the stems; and made zi and chou to name the months, calling them the branches. Stems and branches were then paired to form the sixty-day cycle.）

The phrase "the Yellow Emperor charged Da Rao" in that sentence is legend, not history; it is safer to read it with the two words "it is said" prefixed.

The *Shuowen jiezi* glosses gan as 「筑墙端木也」 ("a pole set upright at the end of a wall under construction"), the post planted at each end when building a wall; it glosses zhi as 「去竹之枝也」 ("taking a branch from a bamboo"), the act of detaching a bamboo branch from the bamboo.

One is the main trunk, the other a side branch. Which is principal and which subordinate is fixed by the names themselves.

The *Erya*, "Explaining Heaven", preserves an earlier set of names: the heavenly stems were called "year yang", the earthly branches "year yin". The names are out of use today, but they show that before the stems and branches became symbols they were first a way of giving names to time.

2. Ten characters, the life of one plant

Which ten characters are the ten heavenly stems?

Jia, yi, bing, ding, wu, ji, geng, xin, ren, gui.

Read in order, one after another, they tell the whole course of a plant from cracking its shell to setting its seed, and then to the seed's return to storage.

Jia (甲) is breaking out of the shell. The *Shiji*, "Treatise on Pitch Pipes", writes 「万物剖符甲而出也」 ("all things split their shell and come forth"); the *Shuowen jiezi* says 「从木，戴孚甲之象」 ("it follows the tree, with the appearance of bearing a shell"), pushing out with a shell on its head.

Yi (乙) is the crooked first growth. *Shiji*, "Treatise on Pitch Pipes": 「万物生轧轧也。」 ("All things are born with a creaking.") The *Shuowen jiezi* puts it vividly: 「象春草木冤曲而出，阴气尚强，其出乙乙也。」 ("Like the spring grasses and trees, which come out crooked and bent, the yin qi still strong, their coming out halting and slow.") The yin qi still presses down; the tip of the shoot can only push out a little at a time.

Bing (丙) is growing stout and showing form. *Shiji*, "Treatise on Pitch Pipes" says 「阳道著明」 ("the way of yang becomes bright"); the *Shuowen jiezi* says 「万物成炳然」 ("all things become brilliant"). The shoot has opened out and the colour brightens.

Ding (丁) is growing firm and taking shape. *Shiji*, "Treatise on Pitch Pipes": 「万物之丁壮也。」 ("The prime of all things.")

Wu (戊) is luxuriance and density. The *Shiming*: 「戊，茂也。」 ("Wu means luxuriant.")

Ji (己) is having form and being recordable. The *Shiming* says plainly: 「己，纪也，皆有定形可记识也。」 ("Ji means to record; all have a fixed form and can be recorded and known.") The form is fixed and can be noted.

Geng (庚) is firm and drawing inward. The *Shuowen jiezi*: 「象秋时万物庚庚有实也。」 ("Like the autumn, when all things are geng geng and solid.")

Xin (辛) is at the height of abundance and coming under constraint. The *Shuowen jiezi*: 「秋时万物成而熟。」 ("In autumn all things ripen and mature.") At the time of harvest they also begin to be bound by the climate.

Ren (壬) is conceiving and nourishing. *Shiji*, "Treatise on Pitch Pipes": 「壬之为言任也，阳气任养万物于下也。」 ("Ren means to bear; the yang qi bears and nourishes all things below.") The new life is already conceived beneath.

Gui (癸) is closing and storing, awaiting sprouting. *Shiji*, "Treatise on Pitch Pipes": 「癸之为言揆也，言万物可揆度。」 ("Gui means to measure; it says that all things can be measured.") Put away,

waiting to be measured again and to sprout again.

Read through the ten characters, and the life of a single plant has also run its course.

There is also a set of one-character mnemonics circulating online: jia-chai, yi-ya, bing-bing, ding-qiang, wu-mao, ji-ji, geng-geng, xin-xin, ren-ren, gui-kui. It is easy to remember, but it is a later summary, not the literal text of the Shiji, "Treatise on Pitch Pipes". When quoting it, do not call it the original words of the Shiji.

3. The twelve branches speak of qi coming and going

The twelve earthly branches change the lens.

The heavenly stems speak of a single plant from birth to storage; the branches speak of when qi comes out and when it is pressed back. The Shiji, "Treatise on Pitch Pipes", glosses zi: 「子者滋也，言万物滋于下也。」 ("Zi means to nourish; it says that all things are nourished below.") Zi is in the eleventh month; the yang qi stirs underground.

By wu it has become 「阴阳交，故曰午」 ("yin and yang cross, and so it is called wu"). The Shuowen jiezi is clearer: 「𠄎也。五月，阴气午逆阳，冒地而出。」 ("To oppose. In the fifth month, the yin qi opposes the yang head-on and bursts up from the ground.") The yin qi pushes upward and collides with the yang qi.

Zi and wu are the two pivots of the system. Zi is around the winter solstice, when the yang qi turns back from its deepest point; wu is around the summer solstice, when the yin qi counter-attacks from above.

The Hanshu, "Treatise on Pitch Pipes and the Calendar", has a general outline that strings the twelve characters together:

孳萌于子，纽牙于丑，引达于寅，冒茆于卯，振美于辰，已盛于巳，罅布于午，昧蓂于未，申坚于申，留孰于酉，毕入于戌，该阂于亥。(In zi it multiplies and sprouts; in chou it knits its buds; in yin it draws forth and reaches out; in mao it pushes up its shoots; in chen it rouses its beauty; in si it is already at its height; in wu it puts forth its calyx; in wei it becomes dark and dense; in shen it extends and hardens; in you it stays and ripens; in xu it completes its entry; in hai it is enclosed and shut away.)

From zi, multiplying below, all the way to hai, which gathers all things into the kernel, and back again to zi.

4. Two rules: yin and yang by odd and even, the five phases by direction

The first rule is yin and yang; it is enough to count positions: the 1st, 3rd, 5th, 7th, 9th and 11th in the sequence are yang, the 2nd, 4th, 6th, 8th, 10th and 12th are yin.

The yang stems are jia, bing, wu, geng and ren; the yin stems are yi, ding, ji, xin and gui. The same for the branches: zi, yin, chen, wu, shen and xu are yang branches; chou, mao, si, wei, you and hai are yin branches.

The second rule is the assignment of the five phases. First remember a table; it is called the five phases: jia and yi pair with wood, bing and ding with fire, wu and ji with earth, geng and xin with metal, ren and gui with water. Why the five phases are five states of motion, and why they are ordered this way, is the subject of notes thirty-one and thirty-two.

Heavenly stems	Five phases	Direction
jia, yi	wood	east
bing, ding	fire	south
wu, ji	earth	centre
geng, xin	metal	west
ren, gui	water	north

The branches are assigned differently: two branches to each of the four directions — yin and mao belong to wood, si and wu to fire, shen and you to metal, hai and zi to water; chen, xu, chou and wei are separately assigned to earth and placed at the four corners.

There is a trap here: apply the stems' scheme to the branches, and one begins by taking zi to be wood.

Zi belongs to water. The stems go two by two in a row; the branches take the four cardinal and four corner positions. The two arrangements are not the same thing.

5. Chen, xu, chou and wei: why both numberings are correct

The branches' assignment to the five phases in fact has a second numbering. Both are correct and they do not clash.

One looks at what the character itself belongs to: yin and mao wood, si and wu fire, shen and you metal, hai and zi water, chen, xu, chou and wei earth. This is called the **root qi** (本气).

The other looks at what the group as a whole governs: yin, mao and chen belong to wood; si, wu and wei to fire; shen, you and xu to metal; hai, zi and chou to water; and chen, xu, chou and wei again to earth. Divided by direction, it is called the **division by quarters** (分方).

The difference arises only over the four characters chen, xu, chou and wei. Take chen: chen is the last character of spring; the wood qi has run to its end and the earth qi is beginning to hold sway, so chen itself belongs to earth; but it is still within the span of spring, and counted by group it still falls to the eastern wood.

There is a further point to state plainly, or the two will seem to contradict.

One rod measures qi. The wood qi appears in the month yin, is strongest in the month mao, and in the month chen gives way to earth.

The other rod measures phenomena (候), the visible state of things in the field. Though the first month is already established as yin, the cold is still heavy and the plants and trees do not stir; by the month mao the grass grows and the orioles fly; by the month chen they are luxuriant.

The two rods are half a beat apart: phenomena lags qi by about a month. So "wood is luxuriant only in the month chen" and "the wood qi gives way in the month chen" do not clash — one speaks of the wood in the field, the other of the qi on the board.

That chen, xu, chou and wei are the four special characters has its reason here: they are the points of handover where the qi has already turned and the phenomena have not yet caught up.

6. One character, three duties at once

The branches are closer to the ground than the stems, because they serve in three places at once.

First, marking the months. The first month is established as yin, the second as mao, on through the eleventh month established as zi and the twelfth as chou. This layer was set out fully in the earlier note on the head of the year.

Second, marking the hours. A day has twelve double-hours; the zi double-hour runs from eleven at night to one in the morning, the chou double-hour from one to three. This layer was listed in the earlier note on the ancient day.

Third, marking the years, paired with the zodiac animals. Zi-rat, chou-ox, yin-tiger, mao-rabbit, chen-dragon, si-snake, wu-horse, wei-goat, shen-monkey, you-rooster, xu-dog, hai-pig.

7. Pairing the twelve branches with animals was not yet fixed in the Qin and Han

Finally, something other accounts mostly do not discuss: when was the pairing of the twelve branches with animals fixed?

The earliest systematic record in the transmitted literature is Wang Chong's *Lunheng* of the Eastern Han. The "Wushi" chapter writes:

寅，木也，其禽虎也。戌，土也，其禽犬也……子亦水也，其禽鼠也。午亦火也，其禽马也。(Yin is wood, its animal the tiger. Xu is earth, its animal the dog. … Zi is also water, its animal the rat. Wu is also fire, its animal the horse.)

The "Wushi" chapter lists only eleven items, missing "chen-dragon". "Chen is the dragon" appears in the Lunheng, "Yandu" chapter. Only the two chapters together make up the full twelve.

So is the Lunheng the earliest?

No. Excavated slips and tablets push the date back a good way.

In 1975, a batch of bamboo slips was unearthed from Qin tomb no. 11 at Shuihudi in Yunmeng, Hubei; the tomb's occupant was buried in 217 BCE (the thirtieth year of the First Emperor of Qin). Among them, the "Daozhe" chapter of the Rishu pairs the branches with animals once over: zi-rat, chou-ox, yin-tiger, mao-rabbit, wu-deer, wei-horse, shen-huan, xu-old-sheep, hai-pig.

The Rishu of the Qin slips from Fangmatan in Tianshui, Gansu, unearthed in 1986, is another pairing; the Rishu of the Han slips from Kongjiapo in Suizhou, Hubei, buried in 142 BCE, is another again.

Set the three batches side by side and two things can be seen.

One is the date. The pairing of branches with animals existed at least by the Qin, some four hundred years earlier than the Lunheng.

The other is that it was "not yet fixed". In the Qin slips wu pairs with the deer, not the horse; wei with the horse, not the goat; xu with the old sheep, not the dog. The same set of pairings was still shifting back and forth around the Qin–Han transition.

Chen Lianshan of Peking University has written an article devoted to sorting this out: the branch–animal pairings in these Rishu do not serve the same purpose as the later zodiac; they are used to infer the appearance of a thief and his hiding place, not to mark years. Strictly speaking, they should be called "day animals" (日肖) at that time, not yet "zodiac animals".

By the Lunheng, the sequence of the twelve animals is basically the same as today's. Using it to mark years and to pair with a person's year of birth is later; how much later has no reliable dating conclusion at present.

Closing

The two characters gan zhi take one from the trunk of a tree and one from its bough. The ten heavenly stems are the life of a single plant; the twelve earthly branches are the coming and

going of a single breath of qi. Each of the two sets has its own rule for fixing yin and yang, and each pairs with the five phases and the directions.

With this, the twenty-two symbols may be counted as fully known.

Knowing them fully is only the first step. Why these twenty-two characters are ordered as they are, and how they pull on one another, is three further long stretches still to come.

A&Q

How many heavenly stems and earthly branches are there, and what are they?

There are ten heavenly stems: jia, yi, bing, ding, wu, ji, geng, xin, ren and gui. There are twelve earthly branches: zi, chou, yin, mao, chen, si, wu, wei, shen, you, xu and hai. Together, twenty-two.

Why are the ten heavenly stems ordered jia, yi, bing, ding?

The order tells the life of a single plant: jia breaking its shell, yi the crooked first growth, bing showing its form, ding taking shape, all the way to gui, closing and storing away, waiting to sprout.

Is zi not of the wood phase?

Zi is water. Yin and mao are wood, si and wu fire, shen and you metal, hai and zi water, and chen, xu, chou and wei earth. To take zi for wood is usually to apply to the branches the stems' arrangement of "two characters to one phase".

In which book does the pairing of the twelve earthly branches with the zodiac animals first appear?

The earliest systematic record in the transmitted literature is Wang Chong's *Lunheng* of the Eastern Han; its "Wushi" and "Yandu" chapters together make up the full twelve. But the Qin slips from Shuihudi, the Qin slips from Fangmatan and the Han slips from Kongjiapo are all earlier, pushing the date back at least to the Qin.

Is that pairing in the Qin slips the same as today's?

No. In the "Daozhe" chapter of the *Rishu* from Shuihudi, wu pairs with the deer, wei with the horse and xu with the old sheep. The pairing was not yet fixed then, nor was its use the marking of years.

Note by Chujia Kuang

The twenty-two symbols are in fact only two kinds of thing: one kind shows how a plant grows, the other how a qi moves. Hold on to these two lines, and it serves far better than rote-memorising the five-phase pairings.

About the author

Chujia Kuang (邝楚嘉), pen name Jiayan Yide (嘉言一得), is an independent researcher in Jieyang, Guangdong, devoted to archiving verifiable primary sources and calendar data.

This column writes about calendrics, solar terms, the stems and branches and orientation, and also about what he reads. The pen name 嘉言一得 comes from the 史记: 愚者千虑, 必有一得, "the foolish man's thousand thoughts must yield one gain". Where something is written wrongly, corrections are welcome.

ORCID: 0009-0002-7650-833X | OpenAlex: A5151908354

Declarations

Funding. This research received no external funding.

Competing interests. The author declares no competing interests.

Data availability. No new data were created or analysed in this study.

Language. The original Chinese text is deposited as a companion file to this record (干支: 从一棵树到二十二个字).

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